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Module 3 Assignment: Arson and Explosives

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Lesson A: Combustion and Arson

1. List three ways that an accelerant used by an arsonist may be detected by a fire investigator (3 marks)

2. a Provide examples of a liquid fuel source and a solid fuel source often used by arsonists. (2 marks)

- b. Explain why an arsonist would choose each of these fuel sources. (2 marks)

3. Describe two differences in the behaviour of an adult arsonist and the behaviour of a juvenile arsonist. (4 marks)

4. Describe one motive that is characteristic of young arsonists and describe how their behaviour can be stopped. (2 marks)

Lesson B: Investigating Arson Fires

5. Describe three reasons arson fires are difficult to solve. (3 marks)

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6. Describe three forms of evidence that indicate that a fire was caused by an arsonist. (3 marks)

"The most common accelerant used in arson fires by far is gasoline because it is easy to obtain and does not cause any suspicions when being purchased."

7. a. What can fire investigators use to identify gasoline residue at a fire scene? (1 mark)

b. Identify a laboratory method that can confirm that evidence from a fire scene contains gasoline (1 mark)

8. Why are gas chromatography and mass spectrometry often both used when analyzing unknown evidence from an arson scene? (3 marks)

Lesson C. Explosives

9. Describe a fuse, and explain why some explosive devices require one. (2 marks)

10. Identify and describe a major disadvantage of using both the X-ray machine and the metal detector to detect explosives. (2 marks)

11. Could the EGIS system replace the use of X-ray machines or metal detectors in airports? Explain. (3 marks)

12. Identify three features of the device used to destroy Pan Am flight 103. (3 marks)

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Lesson D. Case Study Involving Arson and Explosives

13. Describe how Raymond Lee Oyler was able to set the wildfires without being seriously burned. (3 marks)

14. What have some suggested could have prevented the death of the five firefighters in the Esperanza Fire? (2 marks)

15. Was the bomb used to destroy the Alfred P. Murrah Federal Building a high explosive or a low explosive, or did it have traits of both? Explain your choice. (3 marks)

16. Michael Fortier knew about the Oklahoma City Bombing, yet he failed to inform authorities. Explain why he received a light sentence despite this. (3 marks)